

Advanced Statistical Methods II

Lecture 16: Quantile Regression II

Monitirtha Dey

BSDS Third Year Students

Indian Statistical Institute



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Outline of Today's Lecture

Recap and roadmap

Why the check loss targets quantiles

First-order and subgradient conditions

Equivariance properties

Consistency and asymptotic normality

Inference

Practical implementation issues

Summary and exercises

$$\hat{\beta}(\tau) = \arg \min_{\mathbf{b} \in \mathbb{R}^p} \sum_{i=1}^n \rho_{\tau}(y_i - \mathbf{x}_i^{\top} \mathbf{b}), \quad \rho_{\tau}(u) = u\{\tau - \mathbb{I}(u < 0)\},$$

Recall: The check loss function



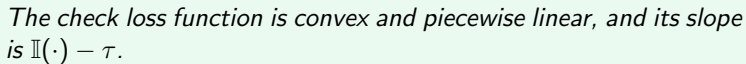
For $0 < \tau < 1$, define the **check loss**

$$\rho_{\tau}(u) = \begin{cases} \tau u, & u \geq 0, \\ (\tau - 1)u, & u < 0. \end{cases}$$

Equivalently,

$$\rho_{\tau}(u) = u\{\tau - \mathbb{I}(u < 0)\}.$$

- ▶ If $\tau = 1/2$, then $\rho_{1/2}(u) = \frac{1}{2}|u|$.
- ▶ If τ is large, positive residuals are penalized more than negative residuals.
- ▶ This asymmetry forces the fitted value upward toward an upper quantile.



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Notation for today

- ▶ Score function: $\psi_\tau(u) = \tau - \mathbb{I}(u < 0)$, so that $\rho_\tau(u) = u \psi_\tau(u)$.
- ▶ Residuals at $\mathbf{b}$: $r_i(\mathbf{b}) = y_i - \mathbf{x}_i^\top \mathbf{b}$.
- ▶ Sample objective: $R_n(\mathbf{b}) = \sum_{i=1}^n \rho_\tau\{r_i(\mathbf{b})\}$.
- ▶ Conditional distribution: $F(y \mid \mathbf{x}) = \mathbb{P}(Y \leq y \mid \mathbf{X} = \mathbf{x})$, density $f(y \mid \mathbf{x})$.
- ▶ True parameter at level τ : $\beta_0 = \beta(\tau)$, errors $u_i = y_i - \mathbf{x}_i^\top \beta_0$.
- ▶ Two matrices that will appear repeatedly:

$$\Sigma = \mathbb{E}(\mathbf{X}\mathbf{X}^\top), \quad \mathbf{J} = \mathbb{E}\{f(\mathbf{X}^\top\beta_0 \mid \mathbf{X})\mathbf{X}\mathbf{X}^\top\}.$$

The **sparsity** (or quantile-density) function is

$$s(\tau) = \frac{d}{d\tau} F^{-1}(\tau) = \frac{1}{f\{F^{-1}(\tau)\}}.$$

A convenient form of the risk



Let $Y \sim F$. Define the (centred) risk

$$R_\tau(a) = \mathbb{E}\{\rho_\tau(Y - a) - \rho_\tau(Y)\}.$$

- ▶ $\rho_\tau(u) = u\{\tau - \mathbb{I}(u < 0)\}$ is Lipschitz with constant $\max(\tau, 1 - \tau) \leq 1$, so $|\rho_\tau(Y - a) - \rho_\tau(Y)| \leq |a|$.
- ▶ Hence $R_\tau(a)$ is finite for **every** distribution F , even with $\mathbb{E}|Y| = \infty$ (e.g. Cauchy).
- ▶ If $\mathbb{E}|Y| < \infty$, then R_τ and $a \mapsto \mathbb{E}\rho_\tau(Y - a)$ differ by a constant, so they have the same minimizers.

Quantiles exist for every distribution; subtracting $\rho_\tau(Y)$ lets the optimization characterization hold equally generally.

The key identity



Lemma 1

For all $y, a \in \mathbb{R}$,

$$\rho_\tau(y - a) - \rho_\tau(y) = \int_0^a \{\mathbb{I}(y \leq t) - \tau\} dt.$$

Consequently $R_\tau(a) = \int_0^a \{F(t) - \tau\} dt.$

Proof. The map $t \mapsto \rho_\tau(y - t)$ is continuous and piecewise linear, with slope

$$\frac{d}{dt} \rho_\tau(y - t) = -\psi_\tau(y - t) = \mathbb{I}(y < t) - \tau, \quad t \neq y.$$

By the fundamental theorem of calculus,

$\rho_\tau(y - a) - \rho_\tau(y) = \int_0^a \{\mathbb{I}(y < t) - \tau\} dt$, and $\mathbb{I}(y < t) = \mathbb{I}(y \leq t)$ except at the single point $t = y$. Taking expectations and using Fubini (the integrand is bounded on a bounded interval) gives $\int_0^a \{F(t) - \tau\} dt$. $\square$

Quantiles are exactly the minimizers



Theorem 1

R_τ is convex, with right and left derivatives

$$R'_\tau(a+) = F(a) - \tau, \quad R'_\tau(a-) = F(a-) - \tau.$$

Hence $\arg \min_a R_\tau(a) = \{a : F(a-) \leq \tau \leq F(a)\}$, the set of τ th quantiles.

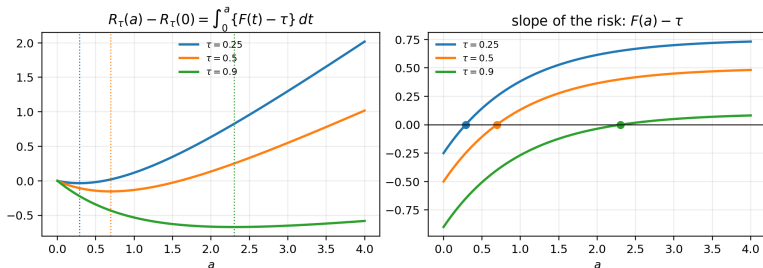
Proof.

- ▶ $t \mapsto F(t) - \tau$ is nondecreasing, and the integral of a nondecreasing function is convex.
- ▶ $\frac{1}{h} \int_a^{a+h} \{F(t) - \tau\} dt \rightarrow F(a) - \tau$ as $h \downarrow 0$ by right-continuity of F ; similarly from the left.
- ▶ A convex function is minimized at a iff $R'_\tau(a-) \leq 0 \leq R'_\tau(a+)$. □

The minimizer set is a nonempty closed interval: a single point unless F is flat at level τ .

Picture: the slope of the risk is $F(a) - \tau$

$Y \sim \text{Exp}(1)$: minimizer is where $F(a) - \tau$ changes sign (dotted: true quantiles)



- ▶ Left: the risk is convex and bottoms out at the true quantile $-\log(1 - \tau)$.
- ▶ Right: its slope $F(a) - \tau$ crosses zero exactly there.

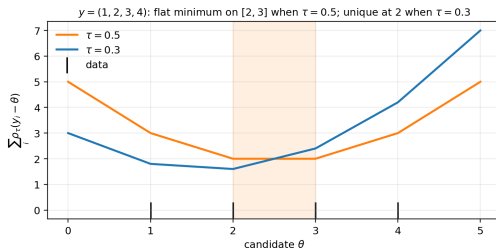
Sample quantiles and non-uniqueness



Applying Theorem 1 to the empirical distribution F_n :

$$\arg \min_{\theta} \sum_{i=1}^n \rho_{\tau}(y_i - \theta) = \{\theta : F_n(\theta^-) \leq \tau \leq F_n(\theta)\}.$$

- ▶ If $n\tau \notin \mathbb{Z}$: unique minimizer $y_{(\lceil n\tau \rceil)}$.
- ▶ If $n\tau \in \mathbb{Z}$: every point of $[y_{(n\tau)}, y_{(n\tau+1)}]$ is a minimizer.



Theorem 2

Among all measurable functions g ,

$$Q_Y(\tau \mid \cdot) \in \arg \min_g \mathbb{E}\{\rho_\tau(Y - g(\mathbf{X})) - \rho_\tau(Y)\}.$$

Proof. Condition on $\mathbf{X} = \mathbf{x}$. By Theorem 1 applied to $F(\cdot \mid \mathbf{x})$, the inner expectation $\mathbb{E}\{\rho_\tau(Y - a) - \rho_\tau(Y) \mid \mathbf{X} = \mathbf{x}\}$ is minimized at $a = Q_Y(\tau \mid \mathbf{x})$. Minimize pointwise, then average over $\mathbf{X}$. $\square$

Subgradients of the check loss

ρ_τ is convex but has a kink at 0. Its **subdifferential** is

$$\partial \rho_\tau(u) = \begin{cases} \{\tau\}, & u > 0, \\ [\tau - 1, \tau], & u = 0, \\ \{\tau - 1\}, & u < 0. \end{cases}$$

- ▶ For $u \neq 0$, this is the ordinary derivative $\psi_\tau(u)$.
- ▶ At the kink, every slope between the left and right derivatives is a subgradient.
- ▶ For convex h : $\mathbf{b}^*$ minimizes $h \iff \mathbf{0} \in \partial h(\mathbf{b}^*)$.

Subgradient calculus is the replacement for “set the derivative to zero” when the objective has kinks.

The exact optimality condition



Let $Z = \{i : r_i(\hat{\beta}) = 0\}$. Since $\partial R_n(\mathbf{b}) = -\sum_i \mathbf{x}_i \partial \rho_\tau\{r_i(\mathbf{b})\}$:

Theorem 3 (subgradient condition)

$\hat{\beta}$ minimizes R_n if and only if there exist $v_i \in [\tau - 1, \tau]$, $i \in Z$, such that

$$\sum_{i \notin Z} \mathbf{x}_i \psi_\tau(r_i) + \sum_{i \in Z} \mathbf{x}_i v_i = \mathbf{0}.$$

Directional derivatives

An equivalent, very useful form. For a direction $\mathbf{w} \in \mathbb{R}^p$,

$$\begin{aligned} R'_n(\mathbf{b}; \mathbf{w}) &= \lim_{\delta \downarrow 0} \frac{R_n(\mathbf{b} + \delta \mathbf{w}) - R_n(\mathbf{b})}{\delta} \\ &= - \sum_{i: r_i \neq 0} \mathbf{x}_i^\top \mathbf{w} \psi_\tau(r_i) + \sum_{i: r_i = 0} \rho_\tau(-\mathbf{x}_i^\top \mathbf{w}). \end{aligned}$$

- ▶ Nonzero residuals move linearly with slope $-\mathbf{x}_i^\top \mathbf{w} \psi_\tau(r_i)$.
- ▶ A zero residual becomes $-\delta \mathbf{x}_i^\top \mathbf{w}$, contributing $\rho_\tau(-\delta \mathbf{x}_i^\top \mathbf{w}) = \delta \rho_\tau(-\mathbf{x}_i^\top \mathbf{w})$.

Optimality via directions

$$\hat{\beta} \text{ minimizes } R_n \iff R'_n(\hat{\beta}; \mathbf{w}) \geq 0 \text{ for every } \mathbf{w} \in \mathbb{R}^p.$$

We use this with simple directions $\mathbf{w} = \pm \mathbf{e}_1$ to prove the counting property.

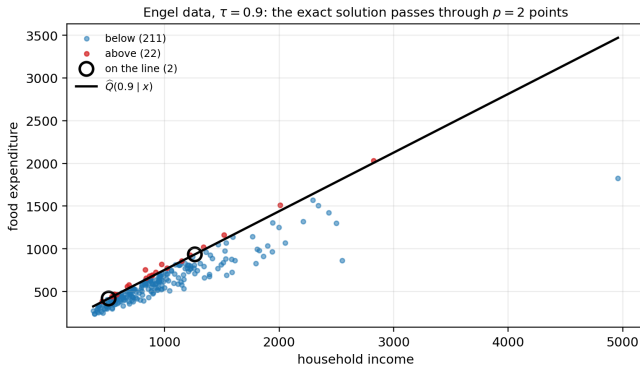
Theorem 4 (Koenker and Bassett, 1978)

If $\mathbf{X}$ has rank p , there is a solution of the form

$$\hat{\beta}(\tau) = \mathbf{X}(h)^{-1} \mathbf{y}(h),$$

for some index set $h \subset \{1, \dots, n\}$ with $|h| = p$ and $\mathbf{X}(h)$ nonsingular. So at least p residuals are exactly zero.

Interpolation on the Engel data



- ▶ $p = 2$ (intercept and income), and exactly two households lie on the fitted 0.9 line.
- ▶ 211 households lie below, and $211/235 \approx 0.898$.

The counting property



Let N_- , N_0 , N_+ be the numbers of negative, zero, and positive residuals at a solution.

Theorem 5 (counting property)

If $\mathbf{x}_i$ contains an intercept, then

$$N_- \leq n\tau \leq N_- + N_0.$$

Proof. Take $\mathbf{w} = \mathbf{e}_1$ (raise the intercept), so every residual drops by δ :

$$0 \leq R'_n(\hat{\beta}; \mathbf{e}_1) = -\tau N_+ + (1 - \tau)(N_- + N_0) \Rightarrow N_- + N_0 \geq n\tau.$$

Take $\mathbf{w} = -\mathbf{e}_1$ (lower the intercept), so every residual rises by δ :

$$0 \leq R'_n(\hat{\beta}; -\mathbf{e}_1) = \tau(N_+ + N_0) - (1 - \tau)N_- \Rightarrow N_- \leq n\tau.$$

Both steps use $N_- + N_0 + N_+ = n$. □

Counting property: Engel check



Exact LP solutions, $n = 235$, $p = 2$:

τ	$\hat{\beta}_1(\tau)$	N_-	N_0	N_+	$n\tau$
0.10	0.4018	23	2	210	23.50
0.25	0.4741	58	2	175	58.75
0.50	0.5602	117	2	116	117.50
0.75	0.6440	175	2	58	176.25
0.90	0.6863	211	2	22	211.50

- ▶ In every row, $N_- \leq n\tau \leq N_- + N_0$.
- ▶ Hence the fraction below the fitted line lies in $[\tau - p/n, \tau]$.

This is the finite-sample, exact meaning of “ $\hat{\beta}(\tau)$ puts a fraction τ of the data below the fit.” Without an intercept, the guarantee is lost.

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Four equivariance properties



Write $\hat{\beta}(\tau; \mathbf{y}, \mathbf{X})$ for the solution computed from data $(\mathbf{y}, \mathbf{X})$.

Theorem 6 (Koenker and Bassett, 1978)

For $\lambda > 0$, $\gamma \in \mathbb{R}^p$, and nonsingular $p \times p$ matrix $\mathbf{A}$:

- (i) scale: $\hat{\beta}(\tau; \lambda \mathbf{y}, \mathbf{X}) = \lambda \hat{\beta}(\tau; \mathbf{y}, \mathbf{X})$;
- (ii) sign reversal: $\hat{\beta}(\tau; -\mathbf{y}, \mathbf{X}) = -\hat{\beta}(1 - \tau; \mathbf{y}, \mathbf{X})$;
- (iii) regression shift: $\hat{\beta}(\tau; \mathbf{y} + \mathbf{X}\gamma, \mathbf{X}) = \hat{\beta}(\tau; \mathbf{y}, \mathbf{X}) + \gamma$;
- (iv) reparametrization: $\hat{\beta}(\tau; \mathbf{y}, \mathbf{XA}) = \mathbf{A}^{-1} \hat{\beta}(\tau; \mathbf{y}, \mathbf{X})$.

When the solution is not unique, read each identity as an equality of solution sets.

Proofs: two facts about ρ_τ



Fact A. $\rho_\tau(\lambda u) = \lambda \rho_\tau(u)$ for $\lambda \geq 0$.

Fact B. $\rho_\tau(-u) = \rho_{1-\tau}(u)$. For $u \neq 0$:

$$\rho_\tau(-u) = -u\{\tau - \mathbb{I}(u > 0)\} = u\{1 - \tau - \mathbb{I}(u < 0)\} = \rho_{1-\tau}(u).$$

Each property is a change of variables in the objective:

- ▶ (i) $\sum \rho_\tau(\lambda y_i - \mathbf{x}_i^\top \mathbf{b}) = \lambda \sum \rho_\tau(y_i - \mathbf{x}_i^\top \mathbf{b}/\lambda)$; put $\mathbf{c} = \mathbf{b}/\lambda$.
- ▶ (ii) $\sum \rho_\tau(-y_i - \mathbf{x}_i^\top \mathbf{b}) = \sum \rho_{1-\tau}(y_i - \mathbf{x}_i^\top (-\mathbf{b}))$; put $\mathbf{c} = -\mathbf{b}$.
- ▶ (iii) $\sum \rho_\tau(y_i + \mathbf{x}_i^\top \boldsymbol{\gamma} - \mathbf{x}_i^\top \mathbf{b}) = \sum \rho_\tau(y_i - \mathbf{x}_i^\top (\mathbf{b} - \boldsymbol{\gamma}))$; put $\mathbf{c} = \mathbf{b} - \boldsymbol{\gamma}$.
- ▶ (iv) $\sum \rho_\tau(y_i - \mathbf{x}_i^\top \mathbf{A}\mathbf{b})$; put $\mathbf{c} = \mathbf{A}\mathbf{b}$.

In each case $\mathbf{b} \mapsto \mathbf{c}$ is a bijection, so minimizers correspond. $\square$

Equivariance: numerical check on Engel



$\tau = 0.25$, exact LP fits:

Data used	$\hat{\beta}_0$	$\hat{\beta}_1$
original ($\mathbf{y}, \mathbf{X}$)	95.484	0.4741
(i) $10 \mathbf{y}$	954.835	4.7410
(ii) $-\mathbf{y}$, fitted at $\tau = 0.75$	-95.484	-0.4741
(iii) $\mathbf{y} + \mathbf{X}\gamma$, $\gamma = (100, 0.2)^\top$	195.484	0.6741
(iv) income measured in thousands	95.484	474.1032

- ▶ Units of y or x do not affect conclusions.
- ▶ (ii) says the lower tail of Y is the upper tail of $-Y$.
- ▶ OLS enjoys (i), (iii), (iv) too. The next property is where QR differs.

Theorem 7

If h is nondecreasing and left-continuous, then

$$Q_{h(Y)}(\tau \mid \mathbf{x}) = h\{Q_Y(\tau \mid \mathbf{x})\}.$$

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Proof (strictly increasing, continuous h).
 $\mathbb{P}\{h(Y) \leq h(q) \mid \mathbf{x}\} = \mathbb{P}(Y \leq q \mid \mathbf{x})$, so $h(q)$ satisfies the defining property of the τ th conditional quantile of $h(Y)$ exactly when q does for Y . $\square$

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- ▶ Means do *not* behave this way: $\mathbb{E}\{\log Y \mid \mathbf{x}\} \neq \log \mathbb{E}(Y \mid \mathbf{x})$ in general (Proof: apply Jensen's inequality!).
 - ▶ So: fit $Q_{\log Y}(\tau \mid \mathbf{x}) = \mathbf{x}^\top \boldsymbol{\beta}(\tau)$, then $\exp\{\mathbf{x}^\top \hat{\boldsymbol{\beta}}(\tau)\}$ estimates $Q_Y(\tau \mid \mathbf{x})$ directly, with no retransformation (smearing) correction.

$(y_i, \mathbf{x}_i)$, $i = 1, \dots, n$, are i.i.d. copies of $(Y, \mathbf{X})$, and $Q_Y(\tau | \mathbf{x}) = \mathbf{x}^\top \beta_0$. So $\mathbb{P}(u_i \leq 0 | \mathbf{x}_i) = \tau$ for $u_i = y_i - \mathbf{x}_i^\top \beta_0$.

- A1.** $\mathbb{E}\|\mathbf{X}\|^2 < \infty$ and $\Sigma = \mathbb{E}(\mathbf{X}\mathbf{X}^\top)$ is positive definite.
- A2.** $F(\cdot | \mathbf{x})$ has a density $f(\cdot | \mathbf{x})$ that is continuous at $\mathbf{x}^\top \beta_0$ and uniformly bounded.
- A3.** $\mathbf{J} = \mathbb{E}\{f(\mathbf{X}^\top \beta_0 | \mathbf{X}) \mathbf{X}\mathbf{X}^\top\}$ is positive definite.
 - ▶ A2 and A3 are local: they only concern f near the target quantile.
 - ▶ No moments of Y are assumed.
 - ▶ A3 fails if the density vanishes at the quantile (e.g. a gap in the support).

See [Koenker \(2005, Ch. 4\)](#) for weaker conditions and fixed-design versions.

Tool 1: Knight's identity



Lemma 2 (Knight, 1998)

For all $u, v \in \mathbb{R}$,

$$\rho_{\tau}(u - v) - \rho_{\tau}(u) = -v \psi_{\tau}(u) + \int_0^v \{\mathbb{I}(u \leq s) - \mathbb{I}(u \leq 0)\} ds.$$

Proof. This is Lemma 1 with $(y, a) = (u, v)$:

$$\int_0^v \{\mathbb{I}(u \leq s) - \tau\} ds = \int_0^v \{\mathbb{I}(u \leq s) - \mathbb{I}(u \leq 0)\} ds + v\{\mathbb{I}(u \leq 0) - \tau\},$$

and $\mathbb{I}(u \leq 0) - \tau = -\psi_{\tau}(u)$ for $u \neq 0$. □

- ▶ The first term is **linear** in v , with a mean-zero random slope.
- ▶ The second term is nonnegative and behaves like a **quadratic** in v after taking expectations. Linear term $\rightarrow$ CLT; quadratic term $\rightarrow$ LLN.

Tool 2: convexity lemmas



The objective has no second derivative, so the usual Taylor-expansion argument does not apply. Convexity replaces it.

Lemma 3 (Pollard, 1991; Hjort and Pollard, 1993)

Let M_n be random convex functions on $\mathbb{R}^p$.

- (a) If $M_n(\mathbf{b}) \rightarrow M(\mathbf{b})$ in probability for each $\mathbf{b}$, then the convergence is uniform on compact sets.
- (b) If $M_n \rightarrow M$ in finite-dimensional distributions, and M has a unique minimizer, then any minimizer of M_n converges in distribution to that minimizer.

- ▶ Part (a) is a probabilistic version of a classical fact for convex functions ([Rockafellar, 1970](#), Thm. 10.8).
- ▶ We only need *pointwise* limits: convexity upgrades them for free.

Theorem 8 (Koenker and Bassett, 1978; Pollard, 1991; Knight, 1998)

Under A1–A3,

$$\sqrt{n}\{\hat{\beta}(\tau) - \beta(\tau)\} \xrightarrow{d} \mathcal{N}(\mathbf{0}, \tau(1 - \tau) \mathbf{J}^{-1} \Sigma \mathbf{J}^{-1}).$$

Moreover, we have the **Bahadur representation**

$$\sqrt{n}\{\hat{\beta}(\tau) - \beta(\tau)\} = \mathbf{J}^{-1} \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_i \psi_{\tau}(u_i) + o_p(1).$$

- ▶ You don't need to remember the exact asymptotic distribution. The main point here is that the Bahadur representation says $\hat{\beta}(\tau)$ is asymptotically a *linear* statistic, just as OLS is.

(a) Sample quantile ($p = 1$, $x_i \equiv 1$): $J = f(q_\tau)$ and $\Sigma = 1$, so the variance is $\tau(1 - \tau)/f(q_\tau)^2$, as in Lecture 15.

(b) Location-shift model $y_i = x_i^\top \alpha + u_i$, u_i independent of x_i . Then $f(x^\top \beta_0 | x) = f_u\{F_u^{-1}(\tau)\} = 1/s(\tau)$ for all x , and

$$\text{Avar} = \tau(1 - \tau) s(\tau)^2 \Sigma^{-1} \quad (\text{OLS: } \sigma^2 \Sigma^{-1}).$$

Only the intercept varies with τ here: $\beta(\tau) = \alpha + F_u^{-1}(\tau) \mathbf{e}_1$.

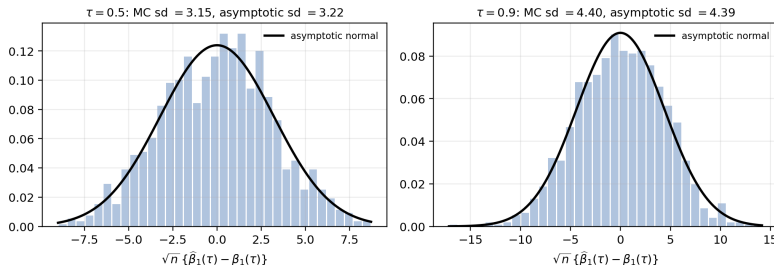
(c) Location-scale model $y_i = x_i^\top \alpha + (x_i^\top \gamma) \varepsilon_i$, $x^\top \gamma > 0$, $\varepsilon_i \sim F_\varepsilon$ independent of x_i . Then

$$\beta(\tau) = \alpha + F_\varepsilon^{-1}(\tau) \gamma, \quad J = f_\varepsilon\{F_\varepsilon^{-1}(\tau)\} \mathbb{E} \left\{ \frac{\mathbf{X} \mathbf{X}^\top}{\mathbf{X}^\top \gamma} \right\}.$$

Slopes change with τ exactly through γ : this is the heteroscedastic picture from Lecture 15.

Does the asymptotic theory work in practice?

Heteroscedastic model, $n = 400$, 1000 replications



Model: $y = 1 + 2x + (1 + 0.5x)\varepsilon$, with $x \sim U(0, 2)$ and $\varepsilon \sim \mathcal{N}(0, 1)$. By case (c), $\beta_1(\tau) = 2 + 0.5 \Phi^{-1}(\tau)$ and $\mathbf{J} = \phi(\Phi^{-1}(\tau)) \mathbb{E}\{\mathbf{X}\mathbf{X}^\top / (1 + 0.5X)\}$.

- ▶ The Monte Carlo spread matches the theoretical sandwich closely at $n = 400$.
- ▶ The $\tau = 0.9$ estimator is noticeably more variable than the median estimator.

Efficiency: median versus mean regression



In the location-shift model, the asymptotic relative efficiency of LAD ($\tau = 0.5$) to OLS is

$$\text{ARE} = \frac{\sigma^2}{1/\{4f_u(0)^2\}} = 4f_u(0)^2\sigma^2, \quad (\text{errors centred at median } 0).$$

Error law	Normal	Logistic	t_5	t_3	Laplace
ARE (LAD vs OLS)	0.637	0.822	0.961	1.621	2.000

- ▶ Under normal errors, LAD needs about $1/0.637 \approx 1.57$ times as many observations as OLS for the same precision.
- ▶ With heavier tails, LAD wins. For Cauchy errors, OLS is not even consistent, while LAD is fine.
- ▶ This is the *Gaussian hare and Laplacian tortoise* trade-off ([Portnoy and Koenker, 1997](#)).

What must be estimated?

$$\widehat{\text{Var}}\{\widehat{\beta}(\tau)\} = \frac{\tau(1-\tau)}{n} \widehat{\mathbf{J}}^{-1} \widehat{\Sigma} \widehat{\mathbf{J}}^{-1}, \quad \widehat{\Sigma} = \frac{1}{n} \mathbf{X}^\top \mathbf{X}.$$

- ▶ $\widehat{\Sigma}$ is easy.
- ▶ $\mathbf{J}$ requires the conditional density **at the quantile**. This is a nonparametric quantity, and it is the crux of QR inference.

Main approaches:

1. **iid**: assume a location-shift model and estimate a single sparsity $s(\tau)$.
2. **nid**: a local density for each observation, from fits at $\tau \pm h$.
3. **Kernel**: smooth the residual density near zero.
4. **Bootstrap**: resample and refit.

- The *Laplacian tortoise* is now competitive with the *Gaussian hare*: with interior-point methods, QR on millions of observations is routine (Portnoy and Koenker, 1997).

A practical checklist



1. **Include an intercept**, which gives the counting property and a clean interpretation.
2. **Choose τ by the scientific question**, then show the full slope path with bands.
3. **Avoid extreme τ with small n** : about $n \min(\tau, 1 - \tau)$ observations inform the fit.
4. **Use robust standard errors** (nid, kernel, bootstrap). Use iid only under a location-shift model.
5. **Check leverage**: QR is robust to vertical outliers, not to outlying x_i .
6. **Transform freely**: by monotone equivariance, you can fit on the log scale and back-transform.

Key takeaway messages



1. $\mathbb{E}\{\rho_\tau(Y - a) - \rho_\tau(Y)\} = \int_0^a \{F(t) - \tau\} dt$: the risk's slope is $F(a) - \tau$, so its minimizers are exactly the τ th quantiles.
2. At a solution, p observations are fitted exactly, and the counting property $N_- \leq n\tau \leq N_- + N_0$ holds.
3. The fit depends on y only through residual signs, which explains robustness to vertical outliers.
4. QR is equivariant to scale, shift, and reparametrization, and to **monotone transformations** of Y .
5. Inference hinges on the density at the quantile. Prefer robust standard errors under heteroscedasticity.

Convexity does the work that differentiability does for least squares.

- ▶ **Censored quantile regression:** survival and duration data.
- ▶ **Nonparametric and additive QR:** $Q_Y(\tau | x) = g_\tau(x)$ via splines or local polynomials.
- ▶ **Penalized QR:** ℓ_1 -penalized QR for many covariates. The penalty keeps the problem a linear program.
- ▶ **Quantile treatment effects:** causal questions about distributions, not only means.

Standard reference for all of the above: [Koenker \(2005\)](#); for a friendly overview of applications: [Koenker and Hallock \(2001\)](#).

[None of these four topics is a part of our syllabus for ASM II.]

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